

David Shumway

Software Engineer | Machine Learning | Backend & Data Systems | Remote
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Senior Software Engineer and Machine Learning Engineer with 10+ years of experience building production backend systems and applied ML pipelines. Master's graduate (CS, UIC) specializing in knowledge graphs, domain adaptation, and transfer learning for environmental and biomedical data. Track record of end-to-end project ownership, from software design and machine learning model development and deployment, to distributed data pipelines, ETL/ELT orchestration, and monitoring, across both research and client-facing environments.

TECHNICAL SKILLS

ML / AI	PyTorch, TensorFlow, scikit-learn, XGBoost, LSTM, GRU, GNN, BERT, CNNs, Random Forest, SVM, domain adaptation (CORAL, ADAPT, EasyAdapt), transfer learning, node embeddings, knowledge graph completion
Data & ETL	Python (Pandas, NumPy, GeoPandas), SQL, PySpark, ETL pipelines, data warehousing
Databases	PostgreSQL, MySQL, Snowflake, Neo4j (Cypher), SQL Server, RDF/SPARQL, OWL
Cloud / DevOps	AWS (RDS, S3, Lambda, ECS, EKS, CloudWatch, Route 53, IAM, Secrets Manager, Cognito), Azure (SQL, Key Vault, Fabric, Data Lake), Docker, CI/CD, OpenStack, MassOpenCloud, CloudLab, Node.js, Next.js, serverless,
Scraping / Retrieval	Scrapy, Python requests, PHP/curl, curl/wget, browser automation (Chrome Extension API, Firefox Extensions, Greasemonkey, userscripts), n8n, tesseract
Ontologies	OWL, RDF, SPARQL, Protégé, RDFLib, knowledge graph design and querying
Visualization	D3.js, ECharts, matplotlib, PowerBI, Tableau, Leaflet
Web-Based	JavaScript, TypeScript, HTML/CSS, Bootstrap, Django, Flask, FastAPI, React, templating (Django/Jinja), custom JS libraries, SQLAlchemy, REST APIs
Languages	Python, SQL, JavaScript, Java, PHP, R, C#, C/C++, Cordova, Android, iOS/Swift

PROFESSIONAL EXPERIENCE

Graduate Research Assistant (ML / Knowledge Graphs) | University of Illinois at Chicago 2021-5/2026

- Lead developer on ONR-funded project (#00014-21-1-2286) applying ML to predict risk to Navy divers at coastal locations.
- Built and compared supervised, semi-supervised, and unsupervised models (LSTM, GRU, GNN, XGBoost, CNN, MLP, Random Forest) for time-series water quality prediction.
- Implemented domain adaptation and transfer learning methods (Feature Augmentation, Balanced Weighting, CORAL, subspace alignment) to generalize models to data-scarce coastal sites, published at IEEE BigData 2024, ES&T 2025.
- Designed and orchestrated ontology-driven ETL/ELT pipelines integrating environmental datasets (NOAA, EPA, WaterQualityData.us) for knowledge graph construction, published at ACM Web Conference 2023.
- Built OWL ontologies in Protégé for waterborne illness risk assessment; queried with SPARQL and RDF.
- Integrated large-scale public datasets (NOAA, EPA, NDBC, NREL, EN4) to improve coastal water quality ML models.

Graduate Research Assistant (Biomedical ML) | University of Illinois at Chicago 2018-2021

- Built MALDI-DB (NIH #R01GM125943), a community-driven public repository with ML workflows for MALDI-TOF spectra, deployed via Docker/OpenStack on MassOpenCloud (Python, Django, R, PostgreSQL, D3.js).
- Developed CNN-based models for bacteria identification in MALDI-TOF mass spectrometry data.
- Co-authored grant proposal for EarthCube semantic data infrastructure connecting distributed scientific datasets.

Teaching Assistant & Instructor | University of Illinois at Chicago 2018-2024

- Served as Instructor of Record for CS586: Data and Web Semantics, taking over mid-semester covering ontology engineering, RDF, OWL, and semantic web.
- Led discussions and mentored students across 11 semesters in courses including Big Data Mining, Software Design, Languages & Automata, and Mobile Development (groups of 5-250).
- Mentored 4 semesters of NSF-REU undergraduate researchers.

Senior Software Engineer | Scalesology, Inc. (Remote / Chicago) 9/2024-2/2026

- Led backend development for a production CRM platform (insurance client): system design, PostgreSQL RDS on AWS, scalable API layer, and client-facing dashboards.
- Migrated legacy SQL Server procedures to Python/Pandas services backed by Snowflake data warehousing workflows.
- Built interactive analytics dashboards and ETL/ELT data pipelines supporting client analytics workflows across energy, grocery, finance, and education sectors (ECharts, React, PowerBI, Tableau, D3.js).
- Implemented logging and observability for backend services supporting production CRM workflows on AWS.

Summer Intern | École Polytechnique Fédérale de Lausanne, DSIL — Lausanne, Switzerland 6/2017-8/2017

- Enhanced Kamusi.org's Neo4j multilingual lexical knowledge graph by analyzing and integrating new datasets.

Co-Owner & Lead Developer | Finger Lakes Business Systems (Remote) 2010-2018

- Operated distributed data automation platform serving 750+ global clients; 3M+ tasks completed at 99.9% approval rating.
- Engineered multi-layer data retrieval systems spanning browser automation (Chrome Extension API, Greasemonkey, userscripts), HTTP-level scraping (PHP/curl, Python requests, curl/wget), and a custom Scrapy-based pipeline serving an internal API used across 30+ production tasks for targeted web data extraction.
- Wrote 375K+ lines of production code (JavaScript, PHP, Python, SQL) across scraping, data quality, and workflow systems.

Digital Archivist & Technical Specialist | Paul Brunton Philosophic Foundation — Burdett, NY 2010–2014

- Curated 100K+ scanned documents and designed LAMP, JavaScript/HTML, Bash, C#, and VBA solutions.
- Automated scanning processes, resulting in 4x efficiency improvement and saving approximately 2-4 years of manual work.

Prior to 2010, held various roles including freelance web development (PHP/MySQL/JavaScript/HTML/CSS), audio transcription, food delivery, retail, and manual labor while self-teaching programming and building foundational technical skills.

EDUCATION

M.S. in Computer Science (August 2026) | University of Illinois at Chicago 2018–8/2026

- PhD dissertation focus: knowledge graphs, domain adaptation, and transfer learning for environmental ML.
- Advisors: Isabel Cruz (in memoriam), Cornelia Caragea. Completed qualifying exam and all coursework.

B.S. in Computer Science | State University of New York at Plattsburgh 2015–2017

- Dean's List all five semesters. Annual CS Department Academic Excellence Award.

SELECTED PUBLICATIONS

Shumway, D. "Engineering a Quota-Constrained Human-in-the-Loop Data Platform: Marketplace-Aware Architecture over 3M+ Microtasks." Available at SSRN (July 2026): (https://papers.ssrn.com/sol3/papers.cfm?abstract_id=7040318).

Shumway, D. "MALDI-DB: Architecture and Implementation of a Community-Driven Platform for Bacterial Identification via MALDI-TOF Mass Spectrometry." Technical report (March 2026). Published online: (<https://davidshumway.github.io/self/MALDI-DB>).

Elahi, Shumway et al. "Machine Learning, Generalization, and Transfer Learning for Predicting the Exceedance of Fecal Indicator Bacteria Thresholds at Beaches." Environmental Science & Technology 2025.

Elahi, Shumway et al. "Predicting Surface Water Bacteria Levels Using Transfer Learning and Domain Adaptation." IEEE BigData 2024.

Gautam, Shumway et al. "Leveraging Existing Literature and Deep Neural Models to Build a Knowledge Graph Focused on Water Quality." ACM Web Conference 2023.

Shumway et al. "A Domain Adaptation Approach for Predicting Recreational Water Quality at Data-Scarce Coastal Locations." Submitted to Environmental Science & Technology 2023.

KEY PROJECTS

Microbial Risk Prediction for Coastal Waters | ONR-funded 2021–2026

- End-to-end ML system combining ontology-driven ETL/ELT data pipelines, transfer learning, and domain adaptation to predict fecal indicator bacteria concentrations at data-scarce coastal sites.
- Tech: PyTorch, TensorFlow, scikit-learn, ADAPT, LSTM, GNN, BERT, OWL/SPARQL/RDF, Dask, GeoPandas, NOAA/EPA APIs.

Modular Insurance CRM Platform | Scalesology 2025–2026

- Led backend development of a greenfield insurance CRM, migrating legacy workflows to a scalable Python/FastAPI architecture on AWS (250K+ LOC).
- Designed relational schema and managed PostgreSQL RDS infrastructure on AWS (RDS, EC2, S3); built core service modules supporting complex CRM workflows via SQLAlchemy and n8n.
- Collaborated cross-functionally to translate evolving client requirements into backend APIs and data models; proactively flagged and mitigated system-level design risks.
- Leveraged LLMs throughout the development lifecycle, generating PRs (GitHub Copilot) and skeleton code, system design, documentation, unit tests, and FastAPI/SQLAlchemy/Python code.
- Tech: Python, FastAPI, SQLAlchemy, PostgreSQL, AWS (RDS, EC2, S3), n8n, Github Copilot.

MALDI-DB: Biomedical ML Repository | NIH-funded 2018–2021

- Community-driven platform for sharing and executing MALDI-TOF mass spectrometry analysis workflows.
- Designed PostgreSQL schema for 40,000+ bacterial spectra with full experimental metadata.
- Integrated R-based ML pipelines (MALDIquant) for spectral similarity search via cosine scoring and replicate collapsing.
- Tech: Python, Django, PostgreSQL, Docker, D3.js, OpenStack. GitHub: (<https://github.com/davidshumway/malddb>).

AMT Worker-Side Execution Platform | Preprint submitted to SSRN (July 2026) 2010–2018

- Engineered a multi-account, quota-aware distributed execution platform over Amazon Mechanical Turk; 3.3M+ approved HITs, \$185K revenue, 99.98% approval rate across 750+ requesters and 1,100+ task types over 8 years.
- Designed factory-based browser extension architecture (~300K LOC JS) with tab-pipelined execution, preview-mode admission control, and cross-worker disk-cached scraping; achieved effective average hourly wages 3–6× above published worker benchmarks; with MySQL backend storing 100s of millions of records (scraper data, task payloads, completion logs).
- Tech: JavaScript, Chrome Extension API, PHP, MySQL, Python (Scrapy), Git, Chart.js, Google Maps API, Fedora Linux.